

16. To Detect Background Jobs, Launch Non-Blocking Processes, Return Prompt Immediately, Store Job Information, Monitor Execution, Test Background Tasks.

To Create Job Table, Store Process Groups, Track States, Update Status, Remove Completed Jobs, Validate Job Records.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        printf("Background process started. PID = %d\n", getpid());

        sleep(5);

        printf("Background process completed.\n");

        exit(0);
    }

    printf("Parent continues immediately.\n");
    printf("Background PID = %d\n", pid);

    printf("Parent is not waiting immediately.\n");

    sleep(2);

    waitpid(pid, NULL, 0);

    printf("Background job finished.\n");

    return 0;
}
```

```
bindu@Bindusree: ~  
bindu@Bindusree:~$ nano skill16.c  
bindu@Bindusree:~$ gcc skill16.c -o skill16  
bindu@Bindusree:~$ ./skill16  
Parent continues immediately.  
Background PID = 607  
Parent is not waiting immediately.  
Background process started. PID = 607  
|
```